

Lead Enrichment Cost Report: Clay vs. Bright Data + Claude

Project Beacon — LinkedIn enrichment pipeline · What's confirmed, what's modeled, and how to get the real Clay number

Revised 2026-08-24 · v2 — corrects the cost-comparison section of the prior draft. Sources: clay.com pricing/docs, this repository's captured Bright Data & Claude web-search test data, platform.claude.com pricing, Clay UI screenshots.

What changed in this revision

The previous draft's §8 presented Clay's cost-per-lead as a clean, confident figure alongside Bright Data's and Claude's *measured* costs. On review, that Clay figure was never measured — it was computed from an assumed "3 credits/lead" waterfall rate that has no supporting data anywhere in this project. This revision:

1. Visually separates **measured** costs (real API usage, real tokens, real dollars) from **modeled** costs (a scenario built on an assumed credit rate) everywhere they appear together.
2. Flags that the completeness shown in the screenshots (§5) — Experience, Education, Languages, Courses, and Projects, each with dated/detailed sub-fields — very likely required more enrichment columns than the single waterfall the cost model assumes, meaning the real cost for *that* level of completeness is probably higher than the modeled figure, not lower.
3. Adds §9, a concrete walkthrough for pulling Clay's actual credit/action usage for these exact leads, so the modeled figures can be replaced with real ones.
4. Softens the closing verdict: completeness is confirmed; the cost comparison is not, and readers should not conclude Clay is cheaper "at scale" until the real number is in.

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1. Executive summary

Clay's manual proof-of-concept delivered complete LinkedIn profile data, but at a cost the team flagged as too high. This report checks two things against real data rather than assumption: whether the pipeline's existing Bright Data scrape, optionally supplemented by Claude's `web_search` tool, can reach the same completeness for free or cheap, and exactly how Clay is billed for the completeness it does deliver.

Verdict: Bright Data's current scrape and Claude web_search are both structurally limited to shallow, snippet-level LinkedIn data — confirmed against real captured data, not documentation claims — and cannot reproduce what Clay's waterfall returned. That part is settled. What Clay's equivalent completeness actually costs is not yet settled — the industry-standard framework is documented in Section 7, but converting it into a real \$/lead number requires reading Clay's own usage dashboard (Section 9), which has not been done yet for this project.

2. Bright Data — raw scrape, checked against 10 real profiles MEASURED

Source file: `raw_responses_LI3.json` . Checked for two fields: About and Experience (work history).

#	Profile	About length	About content	Experience
1	Tammy Pérez	86 chars	Truncated, ends "..."	null
2	Hani Almarzooq	86 chars	Truncated, ends "..."	null
3	Traduttori RH	68 chars	Truncated, ends "..."	null
4	Erika Weber-Guzman, Ed.D.	89 chars	Truncated, ends "..."	null
5	Hani Almarzooq (dup)	86 chars	Truncated, ends "..."	null
6	Cecilia de Tezanos	90 chars	Truncated, ends "..."	null
7	Linka M.	0	None (no public data)	null
8	Michelle Rosslee	1,276 chars	Full, multi-sentence	5 dated roles, full descriptions
9	Chloe Zhang	0	None (no public data)	null
10	(name missing)	0	None (no public data)	null

6 of 10 profiles truncate to 86–90 characters, every one ending in an ellipsis — a systematic snippet limit in the captured dataset itself, not a parsing bug. **9 of 10** return `null` Experience. Only **1 of 10** is fully complete. The pipeline's parser can work exactly as designed and still return an incomplete profile 90% of the time — the ceiling is the source data.

3. Claude + web_search — tested on 3 real leads MEASURED

Model: Claude Opus 5. Up to 8 `web_search` calls allowed per lead.

Lead	Outcome	Detail	Input tok	Output tok	Searches	Cost
Divya Shyam	Total failure	Every field null; could_not_find_anything: true	66,800	1,700	8	\$0.46
Mathumitha Senthil	Partial	About: null. Experience: 4 roles, dates null, generic filler.	120,900	3,000	8	\$0.76
Alex Anthraper	Partial, self-flagged incomplete	Model states the About text is a "truncated preview"; full text not retrievable — see §4.	75,200	3,200	8	\$0.53
Average			87,633	2,633	8	\$0.58

Why it fails: LinkedIn blocks Claude's `web_fetch` tool outright, so `web_search` only ever sees Google-indexed snippet fragments of a LinkedIn page — never the real page. Same shallow-data ceiling as Bright Data's scrape, hit from a different angle. 1 of 3 leads returned nothing.

4. Same lead, two methods: Alex Anthraper CROSS-VERIFIED

Alex Anthraper was tested via Claude `web_search` (above) and appears in a real Clay UI screenshot (§6). This is not a coincidence in this report — it was checked directly against the underlying raw files: both list the identical profile URL `linkedin.com/in/alex-anthraper`, the identical current title "Sr. Project Coordinator," the identical company "Global3," and the identical location "Bengaluru, Karnataka, India." Same person, same profile, two methods.

Field	Claude Opus 5 + <code>web_search</code>	Clay (screenshot, §6)
Summary / About	Truncated preview only ; model states full text "could not be retrieved"	Full paragraph, not truncated
Experience	Not recovered in test output	7 entries, dated, with company domain, locality, revenue range, hybrid/remote type
Cost for this lead	\$0.53	See §8–9 — not yet measured

This is a direct, same-lead comparison rather than an average across different people — it is the clearest evidence in this repo that the **completeness** gap is structural, not a fluke of which leads were sampled. (The *cost* side of this same comparison is addressed in §8–9 below.)

5. Why the free/cheap path isn't working — summary

- LinkedIn blocks deep access to both pipelines.** Neither Bright Data's scraper nor Claude's `web_fetch` tool can load a full, authenticated LinkedIn profile page.
- Bright Data's captured dataset is itself snippet-limited** — 86–90 character About fields, ellipsis-terminated, on 6 of 10 profiles. A ceiling in the source data, not a bug in the parser.

3. **Claude web_search only sees what Google indexed** — short snippet fragments, never the full page, because LinkedIn returns a block response to web_fetch.
4. Both methods hit the identical ceiling from different angles — one is a static/limited scrape dataset, the other is search-engine snippet text. Neither is a data provider with its own maintained profile database.
5. **Clay's waterfall works because it queries separate third-party data providers** (Apollo, Hunter, Findymail, Clearbit, etc.) — companies that maintain their own sourced/refreshed profile data, rather than scraping or search-indexing LinkedIn live. That is the likely structural reason its output is more complete, and also why it isn't free: every provider query is a paid Data Credit. (This explanation is a reasonable inference from how Clay's waterfall is documented to work, not a per-field provenance check on these two specific leads — Clay's own UI does not show which underlying provider populated which field in the cell-detail view.)

6. Clay UI — what "complete" actually looks like

REAL SCREENSHOTS

Two leads pulled directly from Clay's own enrichment output (cell detail view).

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Cell details

🔒

☐

🔍

Search

T Dob null

T Org Soundtrack Studios

🔗 Url <https://www.linkedin.com/in/avikthetddy>

T Name Avik Teddy Chakraborty

T Slug avikthetddy

T Title Proprietor

📄 Awards null

T Country India

>

📄 Courses [1]

T Summary To develop my career in the field of media, where I will be a valuable team member, contributing quality ideas and work for an organization where growth and prospects are ultimate, deliver to fullest of my capabilities & Technical expertise to become an asset to the Organization.

T Headline A Bengali voice actor with over 15 years of experience and lent voice for #tvc #radiospots #elearning #explainervideos #narration #ivr #audiobooks #corporatevideos #dubbing and many more.

>

📄 Projects [1]

>

📄 Education [2]

>

📄 Languages [3]

T Last Name Chakraborty

▼

📄 Experience [5]

>

📄 0 {13}

▼

📄 1 {13}

🔗 Url null

T Title Voice Over Artist

Org Id null

T Company Media

T Summary • At present I am a Radio Presenter at Community Radio JU 90.8 MHz for last 6 years. • I have lent my voice for National Geographic Channel Bengali, Fox Life Bengali, Discovery Bengali. • I have also lent my voice for TV shows like "Adaalat Bengali", "Aahat Bengali", "Crime Petrol Dastak Bengali", "Crime Petrol Satark Bengali", "Ekela Bengali", "Jokhon Bhut Ashey", "Surya The Super Cop - Bengali", "Encounter Bengali" and others. • I have also lent my voice for commercial TV ads like, Xoxo quit addiction, Ayurwin Nutrilim, Ayurwin Nutrigain, Moshk Agarbatti (Bengali), Supply Medicine Dot Com and Others. • I have lent my voice For Hollywood's Renowned Production House "PARAMOUNT PICTURES". I have work with them for feature films like, Kung Fu Panda -I (Bengali), Kung Fu Panda -II (Bengali) & How To Train Your Dragon (Bengali). • I have lent my voice for a Super Hit American Daily Soap Called "FEAR THE WALKING DEAD", "FEAR THE WALKING DEAD - ii" for the character "Christopher" in Bengali Language. • I have also lent my voice for corporate films. Those corporate companies are "Hindustan Unilever" and "Lafarge Cement". • I have

⚡

Cell details

🔒

☐

T Dob null

T Org Global3

🔗 Url <https://www.linkedin.com/in/alex-anthraper>

T Name Alex Anthraper

T Slug alex-anthraper

T Title Sr. Project Coordinator

📄 Awards null

T Country India

📄 Courses null

T Summary Experienced project manager and multilingual content specialist with 10+ years in media, localization, and client services. Expert in managing full project lifecycles with a focus on quality, efficiency, and client satisfaction. Skilled in localization workflows, subtitling, and cross-team collaboration. Combines creative insight with technical precision to deliver results.

T Headline Project Manager with expertise in Localization and Quality Assurance

📄 Projects null

>

📄 Education [2]

📄 Languages null

T Last Name Anthraper

▼

📄 Experience [7]

▼

📄 0 {13}

🔗 Url <https://www.linkedin.com/company/global3>

T Title Sr. Project Coordinator

Org Id 105600529

T Company Global3

T Summary null

T End Date null

T Locality Bengaluru, Karnataka, India

T Location Type Hybrid

Company Id null

☑ Is Current true

T Start Date 2025-10-01

🔗 Company Domain global3.io

T Yearly Revenue Range 1M-5M

▼

📄 1 {13}

🔗 Url <https://www.linkedin.com/company/absolute-translations-ltd>

T Title Project Manager

Org Id 9347250

T Company Absolute Translations Ltd

T Summary Absolute Translations is a UK-based company offering comprehensive 360° language services. It specializes in legal document translation and subtitling, consistently earning a Trustpilot rating of 4.8. End-to-End Project Oversight: Managed all

also lent my voice for a Bengali Documentary named, "UTTAR DINAJPUR KOTHO KOTHA". • My Upcoming TV Shows On Sony Aath – Aahat Season 6 (Bengali), CID (Bengali). • I have also work with Radio Mirchi 98.3 Fm in a project named "Mirchi Singha Bahini" for 2 months in the year 2010.	
T	End Date null
T	Locality Kolkata, India

Fig. 1 — Avik Teddy Chakraborty, Voice Over Artist.
Experience [5], Education [2], Languages [3], Courses [1], Projects [1] — all populated. Summary is a full multi-sentence narrative, not a snippet. Dob/Awards correctly null — genuinely absent public data, not a scrape failure.

of 4.0. End-to-End Project Oversight: Managed all projects from post-sales to final delivery, ensuring seamless transitions and on-time completions. Resource Coordination & Profit Optimization: Assessed assets and scheduled projects with available resources while negotiating rates to maximize profits. Rigorous Tracking & Timely Delivery: Maintained daily progress tracking to ensure every project was delivered on or ahead of schedule. Quality Assurance & Finalization: Conducted	
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Fig. 2 — Alex Anthraper, Sr. Project Coordinator.
Experience [7] — 7 dated roles, vs. 0 usable roles from Claude web_search on the same person (§4). Each role includes company domain, locality, location type (hybrid), yearly revenue range. Summary/Headline fully populated, no truncation.

Read this alongside §8: getting Experience entries enriched with company domain, locality, hybrid/remote type, and revenue range — plus separately-populated Education, Languages, Courses, and Projects — is a much richer result than a single email/phone waterfall column typically produces. That likely means several distinct Clay enrichment columns ran to produce these two rows, not one 3-credit lookup. Keep that in mind when reading the modeled cost figures in §8.

7. Clay billing model & bring-your-own-key SOURCED FROM CLAY DOCS

	Data Credits	Actions
What it pays for	Each lookup against a third-party provider in a waterfall (Apollo, Hunter, Findymail, Clearbit, etc.) — one credit per provider attempted, hit or miss	Platform operations: an enrichment step, an AI/LLM prompt call, a CRM push, an API request
Baseline unit cost	~\$0.05 each (cheaper via negotiated volume rates — ~\$0.02–0.05 in practice)	<\$0.01 each
Failed lookups	Still billed. A 5-provider waterfall where 4 miss and the 5th hits still consumes 5 credits, not 1.	

Bring-your-own-key (BYOK)

Column type	Clay's managed key	Your own key
AI / LLM columns (OpenAI, Anthropic)	~\$0.01/call	~\$0.003/call (OpenAI, per Clay's FAQ) — Actions charge still applies either way
Data-provider columns (Apollo, Findymail, Prospeo, Debounce, NeverBounce)	Clay Data Credit charged per attempt	Credit bypassed — provider bills your own account directly; ~70–80% reported savings at high volume, but requires your own paid subscription with that provider (e.g. Findymail BYOK needs Clay Starter, \$149/mo min.)

Plan tiers: Free \$0 (100 credits/500 actions) · Launch ~\$167–185/mo (3,000 credits/15,000 actions) · Growth ~\$446–495/mo (6,000 credits/40,000 actions) · Enterprise custom. Source: clay.com/pricing.

8. Cost per lead — measured vs. modeled, side by side

The two tables below are deliberately styled differently. The first is real: actual token counts, actual API pricing, actual dollars spent. The second is a scenario — a reasonable industry framework, but built on an **assumed** credits-per-lead rate that has not been checked against this project's actual Clay usage.

MEASURED

Bright Data & Claude — real usage, real cost

Method	Cost/lead	Cost/1,000 leads	Cost/10,000 leads	Completeness (measured)
Bright Data raw scrape (existing infra)	~\$0*	~\$0*	~\$0*	1/10 complete · 6/10 truncated · 9/10 null Experience
Claude Opus 5 + web_search	\$0.58	\$584	\$5,840	1/3 total failure · 2/3 partial, self-flagged incomplete
Claude Sonnet 5 + web_search (rate applied to same measured tokens)	\$0.28	\$282	\$2,816	Not independently tested — same LinkedIn block applies

*Bright Data figures are marginal cost only — the scrape already runs as existing infrastructure. The real "cost" is 90% of leads coming back incomplete, not a dollar figure.

MODELED — NOT MEASURED

Clay — scenario built on an assumed credit rate

Scenario	Assumption	\$/lead	\$/1,000 leads	\$/10,000 leads
Best case	1 credit/lead (first provider hits)	\$0.03–\$0.06	\$30–\$60	\$300–\$600
"Typical" waterfall	3 credits/lead	\$0.07–\$0.16	\$70–\$160	\$700–\$1,600
Worst case	5 credits/lead	\$0.11–\$0.26	\$110–\$260	\$1,100–\$2,600

None of the rows above are backed by an actual Clay usage number for this project — there is currently no credit-usage file or export anywhere in this repository. The "3 credits/lead" assumption was picked as a generic "typical waterfall" figure, not derived from the two leads shown in §6. Given the breadth of what those two screenshots actually contain (Experience + Education + Languages + Courses + Projects, each with detailed sub-fields), the real credit consumption for *that* level of completeness is plausibly higher than 3 credits/lead — meaning Clay's true cost could sit above this scenario table, not within it. Do not use this table to conclude Clay is or isn't cheaper than Claude web_search until §9 is done.

9. How to get Clay's real number

This replaces the modeled scenario in §8 with an actual measurement. It takes a few minutes in the Clay UI:

1. **Open the exact table used for the POC** — the one containing Avik Teddy Chakraborty and Alex Anthraper (or the 20-lead batch it belongs to).

2. **List every enrichment column that ran on those rows.** Given what's populated (Experience, Education, Languages, Courses, Projects, Summary, Headline), this is very likely more than one column — check each column header for its provider/type, not just the ones you remember adding.
3. **Open Clay's usage panel** (workspace Settings → Usage/Billing, or the table's own run history if available) and read the Data Credits and Actions consumed *specifically by this table*, ideally filtered to the time window the POC ran in.
4. **Divide by the number of leads actually processed** in that run to get a real credits/lead and actions/lead figure. If the usage panel only shows workspace-wide totals, take a usage snapshot immediately before and after re-running the same columns on a small fresh batch (5–10 leads) to isolate this table's consumption.
5. **Repeat on a larger sample** (the ~20-lead POC batch, not just these 2) before trusting the average — per-lead credit cost varies with how many waterfall attempts a given profile needs before a provider hits, so 2 leads is not enough to generalize from.
6. **Recompute §8's Clay table** with the real credits/lead and the real \$/credit your plan pays (managed-key rate, or your negotiated/BYOK rate if applicable) in place of the 1/3/5-credit assumptions.

Once this number exists, it — not the modeled scenario — is what should be compared against the \$0.58/lead (Opus 5) or \$0.28/lead (Sonnet 5) Claude figures in §8 to answer "is Clay actually cheaper at scale."

10. Recommendation

Confirmed: Bright Data / Claude web_search is not a substitute for Clay on About/Experience completeness — both are capped by the same shallow public-page access problem, verified against this project's own captured data, including a same-lead, cross-verified comparison (§4).

Not yet confirmed: whether Clay is cheaper than Claude web_search "at scale." The completeness comparison is settled; the cost comparison in the prior draft of this report was a plausible-looking model, not a measurement, and the specific completeness shown in §6 suggests the real Clay cost may be higher than that model assumed. Do the walkthrough in §9 before making a budget decision on this basis.

BYOK still only helps in two narrow cases regardless of the number that comes out of §9: (1) the spend was driven by an AI/Claygent research column, or (2) you already hold — or will buy — a separate subscription with the specific waterfall provider Clay used for these columns.

Sources: clay.com/pricing, clay.com/waterfall-enrichment, clay.com/faq, university.clay.com, community.clay.com, platform.claude.com/docs — fetched 2026-08-24. Bright Data / Claude web_search findings drawn directly from this repository: `POC's/linkedin_poc/raw_responses_LI3.json`, `enrichment_pipeline/output/claude_websearch_formatted_output.json`, `enrichment_pipeline/scratch_claude_websearch_linkedin.py`. Clay screenshots: Clay UI, cell detail view, extracted from the prior draft of this report for reuse here.